



Mega Giga Tera, Sepia!

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Difficulty: easy

Description

In the world of IT, a Junior coder had his first task!, A mission to convert a string of data units (volumes), such as b, B, KB, MB, GB, TB, Kb, Mb, Gb, and Tb, into an array of bytes (B) and print it. He was warned that many attempted but failed.

Sepia, a talented coder with a sharp mind. knew that "B" stood for **Byte** and "b" for **bit**, and the letters **K**, **M**, **G**, and **T** represented **kilo**, **mega**, **giga**, and **tera**, respectively. These units could appear in both uppercase and lowercase.

Sepia approached the challenge and successfully converted the entire string into an array of bytes and knew that till now no one could solve the challenge as he did, you and I agree that it's just the beginning, yet can you solve the challenge?

Inputs

- You will be given one line containing a single string with a space separated mix of volumes with different units.

Outputs

- The final result is a space separated string of **bytes (B)**, where each element represents a converted volume value from the input.



Constraints

- Respect the order of the input elements, it should be in the same order when converted.
- Make sure to respect the notations, b and B are not the same, kB and kb and so on.
- Lower or Capital case for the other units rather than b and B is not taken into consideration.

Examples

Input	Output
1500Kb 49b 48MB	1536000 6.125 50331648
5mB 1tb 16kB 9lgb	5242880 137438953472 16384 12213813248
13GB 13gb 1b 1B	13958643712 1744830464 0.125 1